Two Approximations Under One Name: Separating Sobolev Line Transfer from Expansion Opacity in Kilonova Ejecta

I. A controlled three-way harness and validity maps with calibrated lanthanide data

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Abstract

The light emitted by a kilonova—the radioactive glow of matter ejected when two neutron stars merge—must pass through a gas containing millions of absorption lines of freshly synthesized heavy elements. Simulating this is so expensive that essentially all kilonova radiative-transfer codes replace the true, frequency-resolved interaction of light with each line by a shortcut built on the *Sobolev approximation*, most commonly in its binned "expansion opacity" form. These are routinely described as one approximation. This paper shows that they are two, with different error budgets, and asks of each: *how large is the error, and where in kilonova parameter space does it matter?* We construct a deliberately small, fully controlled test harness in which the same ejecta model, the same atomic data, and the same level populations are pushed through three independent calculations: an analytic Sobolev prediction, a deterministic frequency-resolved reference solver written for this work, and the Monte Carlo code SEDONA, which implements both a resolved line treatment and the expansion-opacity treatment. The three resolved-physics calculations agree to $\sim 1\%$ once a thermal-emission convention is matched, in every experiment from a single artificial line to a realistic forest of 153 calibrated La II lines from the GSI kilonova atomic database. Against this validated baseline we separate two errors that the literature conflates under a single name. The *expansion-opacity implementation* carries a strength-set error floor, arising from a per-resonance substitution $\tau \to 1 - e^{-\tau}$, that grows from $\sim 3\%$ for weak-line forests to 40–90% for strong-line forests and survives undiminished down to a Doppler width of $1\,\mathrm{km\,s^{-1}}$—essentially the thermal width of lanthanum—so that no refinement of line widths rehabilitates it. The *Sobolev approximation itself*, by contrast, shows no strength floor whatever and is accurate to ≈ 5–8% at kilonova-relevant line widths; its own failure mode is the neglect of finite line width, invisible at narrow widths and reaching +39% only for strong lines at $300\,\mathrm{km\,s^{-1}}$. A sweep of 36 conditions across four wavelength windows, three epochs and three ion mixtures shows this separation to be universal, with the realized maximum line optical depth—not wavelength, epoch or composition—as the controlling variable. In the strong-line, incompletely-blanketed regime where kilonova spectral features form, the shortcut in common use is thus wrong by tens of percent for reasons largely unrelated to the approximation it is named after. We also report a measured cost curve for resolved transport and several methodological findings, including an exact symmetry that makes gradient-based Sobolev validity diagnostics systematically conservative.

Contents

1 Introduction

1.1 Kilonovae in three paragraphs

When two neutron stars spiral into each other and merge, a small fraction of their matter—between roughly a thousandth and a few hundredths of a solar mass—is flung outward at speeds of ten to thirty percent of the speed of light. This ejected matter is one of the very few environments in the universe dense enough in free neutrons to build the heaviest elements (gold, platinum, uranium, and the lanthanide row of the periodic table) through the rapid neutron-capture process, or *r-process*. The radioactive decay of these freshly made nuclei heats the expanding cloud, which then glows for days to weeks: this glow is a *kilonova* (for a review see Metzger, 2020). One has been observed in detail, AT 2017gfo, the optical counterpart of the gravitational-wave event GW170817 (Abbott et al., 2017), whose modelling gave the first direct evidence that such mergers synthesize heavy r-process elements (Kasen et al., 2017).

Everything we want to learn from a kilonova—how much matter was ejected, how fast, and above all *which elements were made*—must be read out of its light. The light escapes only after filtering through the ejecta, and the filtering is dominated by an enormous number of atomic absorption lines. Lanthanide ions are the worst offenders: because of their complex open $4f$ electron shells, a single lanthanide ion can have tens of thousands of energy levels and millions of radiative transitions (Tanaka et al., 2020; Flörs et al., 2026b). Their presence raises the ejecta opacity by orders of magnitude and reddens and lengthens the transient (Barnes and Kasen, 2013). The opacity of the ejecta, and therefore the color, brightness, and spectral features of the kilonova, is controlled by these line forests.

Simulating light transport through millions of lines, each intrinsically narrower than one part per million in frequency, is prohibitively expensive if done directly. Every practical kilonova radiative-transfer code therefore leans on an approximation introduced by Sobolev (1960) for expanding stellar envelopes, usually packaged in the "expansion opacity" form of Karp et al. (1977) and Eastman and Pinto (1993). Section 2 explains both ideas from scratch. The approximation is standard, venerable, and almost never checked against a resolved calculation *under kilonova conditions*. This work begins that check.

It also draws a line the field has left blurred. Sobolev's approximation proper attenuates each line by $e^{-\tau_S}$ and rests on two assumptions: that the gas does not change across a resonance, and that lines do not overlap. Expansion opacity is a further, and separate, step—a statistical device that bins many lines into an effective opacity and, as we show, substitutes $1 - e^{-\tau}$ for τ at every resonance crossing. Codes and papers commonly report using "the Sobolev approximation" when they mean the second. Our central result is that the two carry error budgets differing by roughly an order of magnitude, so the distinction is not pedantry.

1.2 What this paper does

We do not build a kilonova pipeline. We build the smallest apparatus that can measure the two approximation errors cleanly and *separately*, and we operate it on progressively more realistic inputs. Concretely, this first paper reports:

- a **three-way validation harness** (Section 3): the same spherically expanding model, atomic data, and level populations evaluated by (a) analytic Sobolev theory, (b) a purpose-written deterministic frequency-resolved solver, and (c) the Monte Carlo code SEDONA (Kasen et al., 2006) in both its resolved and its expansion-opacity modes;

- **verification experiments** (Section 4) in which the resolved-physics calculations agree at the $\sim 1\%$ level, from one artificial line to a twenty-line ladder, establishing the baseline everything else is measured against;
- **the separation itself** (Section 5.3), the central result: on a forest of 153 calibrated La II lines from the GSI database (Flörs et al., 2026b,a), the Sobolev approximation errs by $+2.4\%$ in band flux while expansion opacity errs by $+44.9\%$, and the two fail through different mechanisms;
- **validity maps** (Sections 5.4–5.5): both errors as functions of line strength, Doppler width, temperature, epoch, wavelength and composition—36 further conditions showing the separation to be universal and controlled by the realized maximum line optical depth alone;
- ten **methodological findings** (F1–F10, collected in Table 1), among them an exact symmetry cancellation that makes gradient-based validity diagnostics conservative (F1); the demonstration that expansion opacity attenuates a single strong line by $e^{-(1-e^{-\tau})}$ rather than $e^{-\tau}$ (F4); that this error is *per-resonance*, so it neither averages away with line count (F5) nor vanishes as line widths shrink (F6); and two conventions—thermal emission (F8) and spectral normalization (Section 5.5)—whose mismatch produces spurious few-percent discrepancies in exactly the cross-code comparisons such a study depends on.

Readers with no radiative-transfer background should read Section 2 first; the appendix derives every formula used in the text at a level that assumes only basic calculus.

2 A compact radiative-transfer primer

2.1 Light through matter: intensity, opacity, optical depth

Consider a beam of light of frequency ν traveling through a gas. Its strength is described by the *specific intensity* I_ν (energy per unit time, area, solid angle, and frequency). Along a path length s, the gas removes energy from the beam (absorption) and adds its own (emission):

$$\frac{\mathrm{d}I_\nu}{\mathrm{d}s} \;=\; -\,\alpha_\nu\, I_\nu \;+\; j_\nu. \tag{1}$$

Here α_ν (units cm^{-1}) is the *absorption coefficient*: $1/\alpha_\nu$ is the average distance a photon travels before being absorbed. The emissivity j_ν describes how much the gas glows. In this work the gas is kept in conditions where its own glow at the relevant frequencies is negligible or exactly modeled (Section 3.2), so absorption is the star of the show.

The dimensionless quantity that controls absorption is the *optical depth*

$$\tau_\nu \;=\; \int \alpha_\nu\, \mathrm{d}s, \tag{2}$$

which counts how many absorption mean-free-paths the beam crosses. If the gas does not glow, the solution of Eq. (1) is simply

$$I_\nu^{\mathrm{out}} \;=\; I_\nu^{\mathrm{in}}\, e^{-\tau_\nu} : \tag{3}$$

a beam crossing optical depth $\tau = 1$ keeps 37% of its light, $\tau = 2$ keeps 14%, $\tau = 5$ keeps 0.7%. Every result in this paper is, at bottom, a statement about which τ gets exponentiated. (The full solution with emission is derived in Appendix A.1.)

2.2 Spectral lines and Doppler width

Atoms absorb light most strongly at the discrete frequencies of their internal transitions—*spectral lines*. A line is not infinitely sharp: thermal motion of the absorbing ions Doppler-shifts each ion's absorption slightly, smearing the line into a profile $\phi(\Delta\nu)$ (normalized so $\int \phi \, d\nu = 1$) whose width corresponds to a velocity

$$v_D \;=\; \sqrt{2k_B T/m_{\rm ion}}, \tag{4}$$

typically $\sim 0.6\,{\rm km\,s^{-1}}$ for lanthanum at kilonova temperatures. Expressed as a fraction of the line frequency this is $v_D/c \sim 2\times10^{-6}$: lines are extraordinarily narrow. The absorption coefficient of one line is

$$\alpha_\nu \;=\; \sigma_{\rm cl}\, f\, n_l\, \phi(\nu-\nu_0), \qquad \sigma_{\rm cl} \equiv \frac{\pi e^2}{m_e c} = 0.02654\ {\rm cm^2\,Hz}, \tag{5}$$

where ν_0 is the line's rest frequency, f its dimensionless *oscillator strength* (how intrinsically strong the transition is), and n_l the number density of ions sitting in the transition's lower energy level. How many ions sit in which level is set, in thermal equilibrium, by the Boltzmann distribution (Appendix A.1).

2.3 Expanding ejecta: why geometry rescues us

Kilonova ejecta expand *homologously*: after a short time every parcel coasts at constant speed, so its position is $r = v\,t$ and equivalently

$$v(r) \;=\; r/t, \tag{6}$$

where t is the time since the merger. Velocity increases linearly outward, like raisins in an expanding loaf.

Now follow a single photon of fixed frequency ν flying through this medium. Each gas parcel it passes sees the photon Doppler-shifted by the local line-of-sight velocity. Because the velocity changes monotonically along the flight path, the photon's frequency *as seen by the gas* sweeps continuously redward. It therefore comes into resonance with any given line at **one specific location** — the point where the local Doppler shift brings the gas-frame frequency onto the line center — and is out of resonance everywhere else. In homologous flow this location is particularly simple: for a photon traveling toward an observer along the z axis, the line-of-sight velocity of the gas at coordinate z is z/t for *every* ray, so the resonance locations are flat planes $z = $ const.

2.4 The Sobolev approximation

Sobolev (1960) realized that this geometry makes the line optical depth computable without resolving the profile at all; Castor (1970) cast the same idea in the escape-probability form used by most modern codes. The photon crosses the narrow resonance region quickly; if the gas properties (n_l, temperature) are constant across that thin region, the integral in Eq. (2) collapses (Appendix A.2) to the *Sobolev optical depth*

$$\boxed{\tau_{\rm S} \;=\; \sigma_{\rm cl}\, f\, n_l(r_{\rm res})\, \lambda_0\, t} \tag{7}$$

where λ_0 is the line's rest wavelength and n_l is evaluated *at the resonance point*. Two assumptions are buried here and both are testable: (i) *locality* — the gas does not change across the resonance region (width $\sim v_D$ in velocity); (ii) *isolation* — no other line's resonance region overlaps this one. A photon crossing k line resonances is attenuated by $e^{-\sum_k \tau_{{\rm S},k}}$.

2.5 Expansion opacity: the workhorse shortcut

Transport codes need an opacity per frequency *bin*, not per line. The *expansion opacity* formalism (Karp et al., 1977; Eastman and Pinto, 1993) assigns to each bin an effective absorption coefficient built from the Sobolev depths of all the lines inside it,

$$\alpha_{\rm bin}^{\rm exp} = \frac{1}{c\,t} \sum_{k\in{\rm bin}} \frac{\nu_k}{\Delta\nu_{\rm bin}} \left(1 - e^{-\tau_{{\rm S},k}}\right). \tag{8}$$

This is enormously convenient and it is what most kilonova (and supernova) codes mean when they say they use "the Sobolev approximation." But notice what it does to a single line: integrating $\alpha^{\rm exp}$ over the path on which that line stays inside one bin gives an effective optical depth of $1 - e^{-\tau_{\rm S}}$, so the transmitted fraction is

$$e^{-(1-e^{-\tau_{\rm S}})} \qquad \text{instead of the true} \qquad e^{-\tau_{\rm S}}. \tag{9}$$

For a weak line ($\tau_{\rm S} \ll 1$) these agree, since $1 - e^{-\tau} \approx \tau$. For a strong line they do not: at $\tau_{\rm S} = 2$ the true transmission is 0.135 while expansion opacity gives 0.421; at $\tau_{\rm S} = 50$, the true transmission is essentially zero while expansion opacity still leaks $e^{-1} = 0.37$ per crossing. Each resonance crossing is "capped" at one effective optical depth, no matter how black the line really is (derivation in Appendix A.4). **This substitution, $\tau \to 1 - e^{-\tau}$ per resonance, is the central object measured in this paper.**

3 Methods: the three-way harness

3.1 Design principle

The danger in comparing two radiative-transfer treatments is that differences can hide anywhere: different atomic data, different level populations, different geometry, different frequency grids. Following the design requirement that only the line-transfer treatment may differ, every experiment in this paper evaluates *one* physical configuration three ways:

Leg	Implementation	Role
Analytic	Sobolev theory, Eq. (7),	prediction
	p-averaged over the source disk (App. A.5)	
Deterministic	`sobolev/formal_transfer.py` (this work):	resolved reference
	frequency-resolved formal solution along rays	
Monte Carlo	SEDONA (Kasen et al., 2006; Roth and Kasen, 2015):	production code
	(a) resolved bound–bound mode	resolved cross-check
	(b) expansion-opacity mode	**the treatment under test**

Agreement of the first three (analytic, deterministic, SEDONA resolved) validates the harness; the offset of the fourth is the measured error, reported as

$$\Delta_{\rm Sob} = \frac{F_{\rm exp} - F_{\rm resolved}}{F_{\rm resolved}}, \tag{10}$$

where F is the band-averaged transmitted flux. $\Delta_{\rm Sob}$ is always formed from the *two* SEDONA *runs*, so Monte Carlo conventions, frequency grids, and normalizations cancel; the deterministic solver serves as an independent check of the resolved leg.

3.2 Geometry and physical conventions

All experiments use a spherical, homologously expanding shell (Eq. 6) around a central blackbody "light bulb" of radius $r_{\rm core}$ and temperature $T_{\rm core}$: the shell imprints absorption lines on the core's smooth continuum, and the emergent spectrum is measured far away. Rays that hit the core start with its blackbody intensity; rays that miss it start dark. Three deliberate simplifications keep the comparison interpretable:

1. **Pure absorption, local thermal emission.** Scattering and fluorescence are off; absorbed energy is re-emitted thermally with source function $S_\nu = B_\nu(T)$ (the Planck function). Shell temperatures are chosen so the shell's own glow in the measured band is negligible ($B_\nu(T_{\rm shell}) \ll B_\nu(T_{\rm core})$); every leg treats the identical source, so any residual emission cancels in $\Delta_{\rm Sob}$.

2. **Fixed temperatures.** Radiative equilibrium is disabled; the gas state never feedbacks on the radiation. Populations are therefore known exactly.

3. **Controlled velocities.** Shell velocities span 1000–3000 $\mathrm{km\,s^{-1}}$ (not the realistic 0.1–0.3c), keeping an $\mathcal{O}(v/c)$ frame ambiguity (Finding F3, Appendix A.6) below the measurement level.

3.3 Population control

SEDONA computes its own thermal-equilibrium level populations from the atomic data file it is given. To guarantee both codes use *exactly* the same populations, we generate SEDONA's atomic-data files directly from the same source data the Python side reads: for the La II experiments, all 472 energy levels of the GSI La II model are written into the file, only the transitions in the studied wavelength window are included as lines, and the ionization potential is set to 10^5 eV so that ionization is impossible and the element sits entirely in the intended ionization stage. Both codes then evaluate the same Boltzmann distribution over the same levels; we verified in SEDONA's output gas state that the free-electron fraction is $\sim 10^{-10}$, i.e. the population control holds.

3.4 The deterministic reference solver

The independent resolved leg is a deliberately small (∼200-line) Python program. It casts parallel rays through the sphere at many impact parameters p, evaluates the frequency-resolved line opacity (Eq. 5) at every step using the local Doppler-shifted frequency, and accumulates the formal solution of Eq. (1) by cumulative optical depth (Appendix A.1). Emergent luminosity is the p-weighted integral of ray intensities. Its correctness rests on four analytic limits enforced as unit tests: a bare core reproduces the exact blackbody-sphere luminosity; a single line's absorption trough reaches $e^{-\tau_{\rm S}}$ in the Sobolev limit; two well-separated lines absorb independently while overlapping ones multiply ($e^{-(\tau_1+\tau_2)}$); and a warm shell partially refills the trough. The full test suite of the accompanying repository (19 tests) passes at every commit reported here.

3.5 A convention that must be matched: thermal emission

Finding F8, recorded because it is a trap for anyone benchmarking a deterministic solver against a Monte Carlo code. Our solver integrates the LTE source term $S_\nu = B_\nu(T_{\rm shell})$ along every ray; SEDONA, run at fixed temperature with radiative equilibrium disabled, deposits absorbed packet energy without re-emitting it. Setting $T_{\text{shell}} \to 0$ in the solver isolates the difference exactly: the single-ion forest moves $0.3538 \to 0.3390$ against SEDONA's 0.3426 ($+3.3\% \to -1.0\%$), and the blend $0.2410 \to 0.2249$ against 0.2277 ($+6.5\% \to -1.2\%$). Like for like, the two independent codes agree at the $\sim 1\%$ Monte Carlo noise floor. The term exceeds the naive $B_\nu(3000)/B_\nu(6000) = 2 \times 10^{-3}$ estimate because the shell subtends nine times the core's projected area, and it grows with saturation. Absolute band fluxes quoted from SEDONA in this configuration are therefore attenuation-only; Δ_{Sob}, a differential between two SEDONA runs sharing one convention, is unaffected.

3.6 Atomic data

Real-line experiments use the GSI Database for Kilonova Radiative Transfer (Flörs et al., 2026b,a). It is one of several recent efforts to supply r-process opacity data (cf. Tanaka et al., 2020, 2025); we adopt it because its wavelengths are calibrated against experiment, which matters when the physical question turns on separations of a few $\mathrm{km\,s^{-1}}$. It provides calibrated energy levels and E1 transitions for singly and doubly ionized lanthanides, with per-level calibration flags (`xmatch` = matched to experiment, `shifted` = calibrated shift, `uncalib`). For La II: 472 levels and 17,743 transitions. Internal consistency of the dataset is excellent: oscillator strengths recomputed from the Einstein A coefficients agree with the tabulated $\log gf$ values to 1.5×10^{-4} across our window, so both codes carry identical line strengths regardless of which column they read.

4 Verification experiments

4.1 Reproducing and breaking the Sobolev limit in isolation

Before any code comparison, the toy problem: one line, constant n_l, homologous slab. The resolved integral (Eq. 2) converges to the analytic τ_{S} (Eq. 7) to machine precision ($E_{\text{Sob}} = 2 \times 10^{-16}$ at 2×10^5 grid points), with monotone convergence under refinement—important because an unconverged grid mimics a physical Sobolev deviation.

Breaking locality on purpose (Fig. 1) produced **Finding F1**: with the population gradient placed *symmetrically* about the resonance point, the error cancels identically—the line profile is even while the leading gradient term is odd, and the natural symmetric choice (tanh centered on resonance) also has vanishing curvature there. The error only appears when the resonance sits off-center on the gradient (we use the point where curvature is large), and then follows a clean ϵ^2 law, at the few-percent level when $\epsilon = v_D/v_{\text{scale}} \gtrsim 0.3$ (Appendix A.3). The practical consequence: widely used validity diagnostics based on the gradient magnitude $G = v_D\,|\mathrm{d}\ln n_l/\mathrm{d}v|$ are *conservative*: the true error depends on where the resonance sits on the gradient, and can be far below the G-based bound.

4.2 One line, three codes

The minimal cross-code experiment uses SEDONA's built-in two-level test atom (one line at 1215.5 Å, $f = 0.665$) in a shell tuned to $\tau_{\text{S}} = 2$. The shell temperature (2000 K) was chosen after an ionization- balance check: at the very low densities that give $\tau_{\text{S}} \sim 1$, a 5000 K gas would be fully ionized, destroying population control; at 2000 K the gas is neutral to one part in 10^{10}.

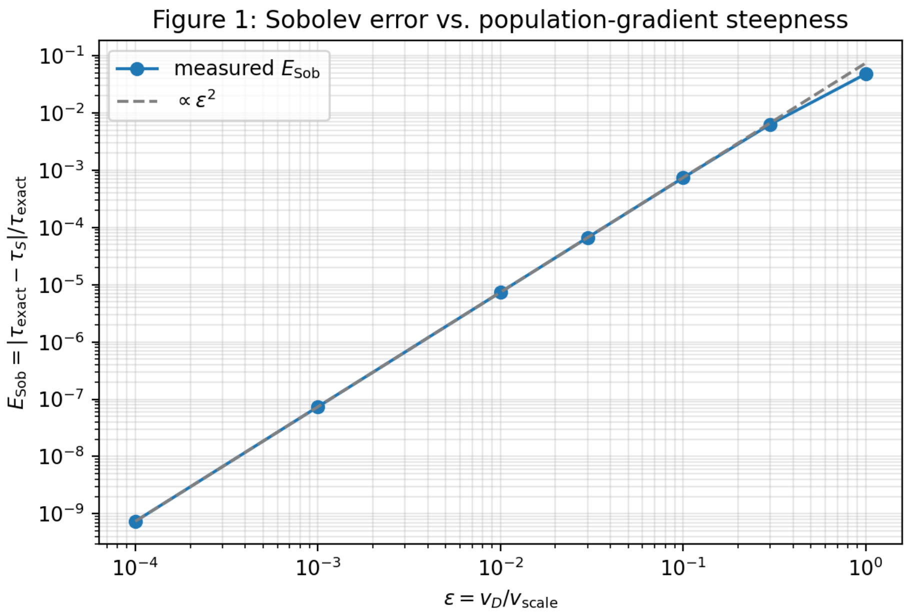

Figure 1: Controlled breakdown of the Sobolev locality assumption. A tanh-shaped population gradient of velocity scale v_{scale} is imposed; the relative error E_{Sob} between the resolved integral and the Sobolev formula follows the predicted ϵ^2 power law in $\epsilon = v_D/v_{\mathrm{scale}}$ (measured log-slope 2.0000), reaching $\sim 5\%$ when the gradient scale approaches the Doppler width.

The measured absorption-trough depths (Fig. 2):

analytic $e^{-\tau_S}$	0.1353
deterministic solver	0.1372
SEDONA resolved	0.1420
SEDONA expansion opacity	**0.4285**
predicted cap $e^{-(1-e^{-\tau_S})}$	0.4212

Two independent resolved implementations—deterministic rays and Monte Carlo—agree with each other and with theory at the few-percent level (**harness validated**), while the expansion-opacity mode lands within 2% of its own predicted cap, a factor 3.2 too much transmitted light. This is **Finding F4**: for a single strong line, "Sobolev mode" as implemented via expansion opacity is *not* Sobolev line transfer.

The same experiment surfaced **Finding F3**: the deterministic solver's 1.4% excess over the analytic value scales with the bulk velocity at the resonance, and at a resonance placed at 7.7% of c grows to 7%, matching $e^{-\tau_S(1-z_{\mathrm{res}}/ct)}$ to four digits. We read this first as a frame ambiguity and then as the leading relativistic correction; both were wrong. Appendix A.6 shows it is an artifact of integrating a *frozen snapshot* of the ejecta: allowing the ejecta age to advance along the photon path, as it physically must, gives $\tau_{\mathrm{rel}}/\tau_S = 1/\gamma$, which has no first-order term at all. The practical consequence is the opposite of what the $\mathcal{O}(v/c)$ reading implied: at our shell velocities the relativistic correction is 5×10^{-5}, so it cannot account for any part of the residual in §5.3.

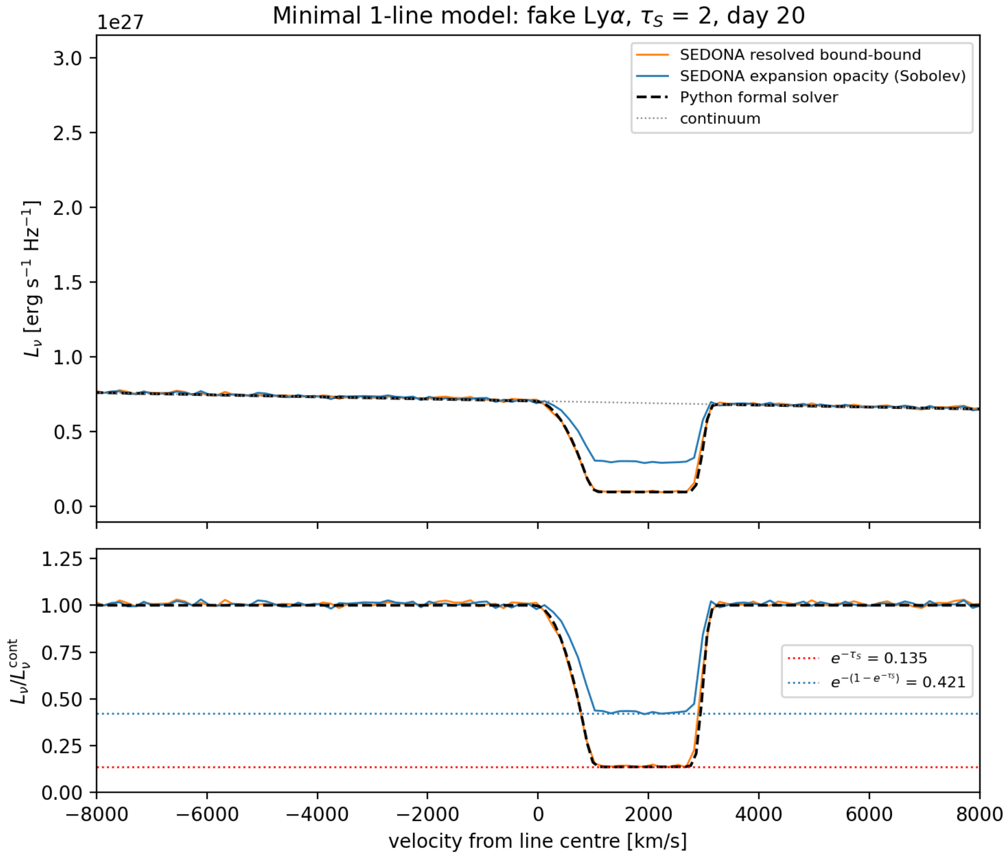

Figure 2: Minimal one-line model (artificial Lyα-like line, $\tau_S = 2$, cold neutral shell). Top: emergent spectra. Bottom: ratio to the continuum. SEDONA's resolved mode (orange) lies on top of the deterministic solver (black dashed) across the entire profile, at the analytic depth $e^{-\tau_S} = 0.135$ (red dotted). The expansion-opacity mode (blue) instead sits at the predicted capped value $e^{-(1-e^{-\tau_S})} = 0.421$ (blue dotted).

4.3 From two lines to a twenty-line ladder

Custom SEDONA atomic files with 2 and then 20 identical-strength lines ($\tau_S = 0.5$ each) test multi-line behavior (Fig. 3). In the center of the 20-line forest, where each photon crosses ~ 4 resonances, five determinations of the transmitted fraction give:

analytic Sobolev (p-averaged)	0.2405
deterministic solver	0.2407
SEDONA resolved	0.2435
analytic expansion cap	0.3194
SEDONA expansion opacity	0.3257

Finding F5: the expansion-opacity error tracks its own per-line cap prediction, i.e. the error is incurred *per resonance crossing*. Adding more lines multiplies both the true and the capped attenuations equally, so the fractional flux error does *not* average away with line count—it only

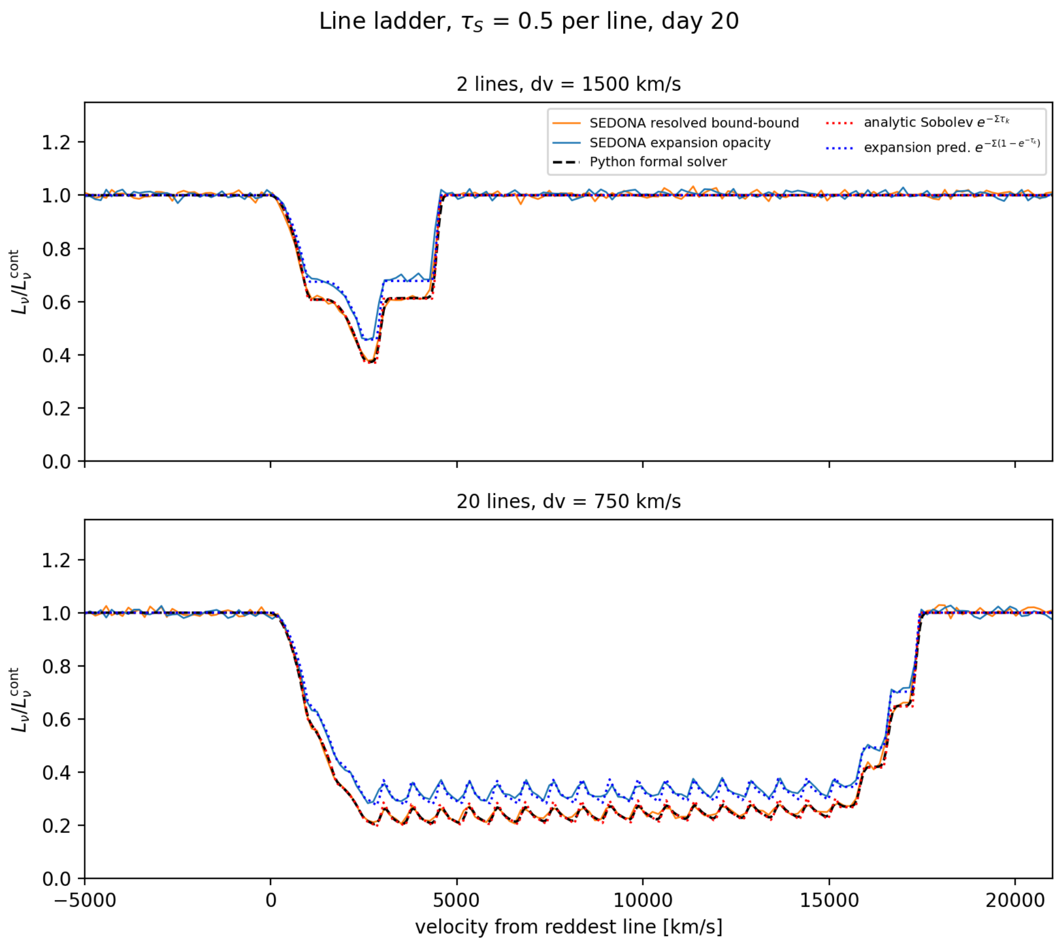

Figure 3: Line ladder at $\tau_S = 0.5$ per line. Top: two lines separated by $1500\,\mathrm{km\,s^{-1}}$ (overlapping troughs). Bottom: twenty lines spaced $750\,\mathrm{km\,s^{-1}}$ (a mini forest). The resolved trio—SEDONA resolved (orange), deterministic solver (black dashed), p-averaged analytic staircase (red dotted)—lock together tooth for tooth; the expansion pair (blue solid: SEDONA; blue dotted: analytic cap prediction) ride systematically above them.

vanishes as $\tau \to 0$ per line. Expansion opacity is a *weak-line* approximation, not a many-line one.

A technical note that matters for anyone reproducing this: the analytic staircase must be averaged over the stellar disk (p-average, Appendix A.5), because resonance planes closer than the core radius are crossed by only the outer rays; naive plane counting predicts visibly too-shallow troughs.

5 Results: two approximations, two errors

With the harness validated, we turn to calibrated La II data. We first ask whether the assumptions underlying either approximation have anything to fear in real line lists (§5.1), then set up a realistic forest (§5.2) and measure both errors on it (§5.3)—the paper's central result. The

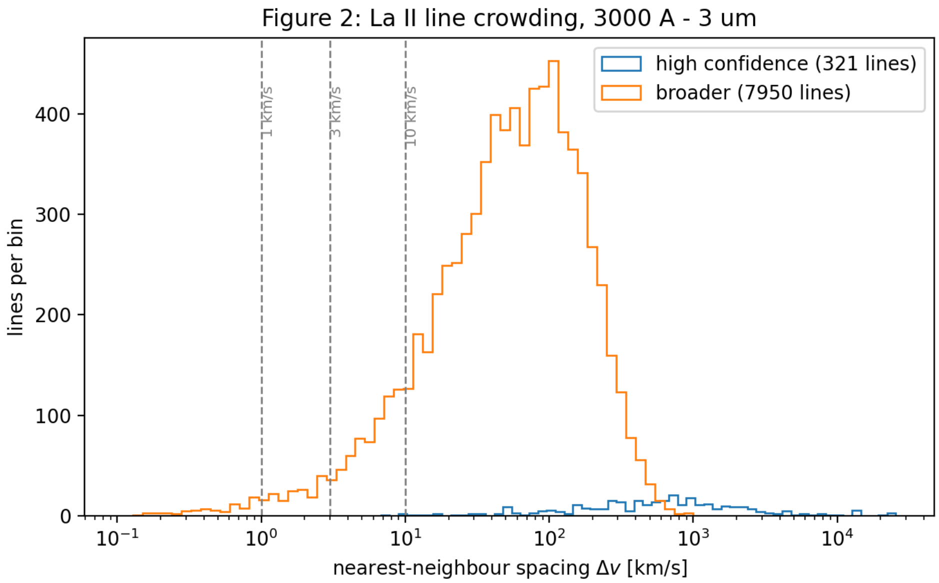

Figure 4: Nearest-neighbor velocity spacing of GSI La II lines (3000 Å–3 μm): in the broader subset, 11.3% of lines have a neighbor within 10 km s^{-1} and 3.3% within 3 km s^{-1}.

remaining subsections establish where each error lives (§5.4), that the picture is not peculiar to one window or epoch (§5.5), how blending changes it (§5.6), and what the shortcut buys in computer time (§5.7).

5.1 Do the assumptions have anything to fear? Line crowding

The Sobolev isolation assumption fails when lines crowd within a Doppler width. Fig. 4 shows that even for this *single* ion, the calibrated data contain a real crowding tail: 11.3% of the broader line sample has a neighbor within 10 km s^{-1}. Multi-ion mixtures only add lines, so the potentially problematic regime exists in nature.

5.2 A realistic forest: 153 calibrated lines

We select the 100 Å window with the richest optical-depth distribution (3850–3950 Å), build the population-controlled SEDONA atom from the GSI data (Section 3.3), and run the three-way comparison at $T = 3000$ K, $t = 1$ day, Doppler width 100 km s^{-1} (Fig. 5). Band-averaged transmitted flux over 3800–3955 Å:

deterministic solver	0.3549
SEDONA resolved	0.3426
SEDONA expansion opacity	0.4965
Δ_{Sob}	**+44.9%**

The resolved pair agree to 3.6% as tabulated—and to 1.0% once the thermal-emission convention is matched (Finding F8, §3.5)—while the expansion-opacity run overestimates the transmitted band flux by 45%. Notably this exceeds the uniform-ladder error (+35% at $\tau = 0.5$ per

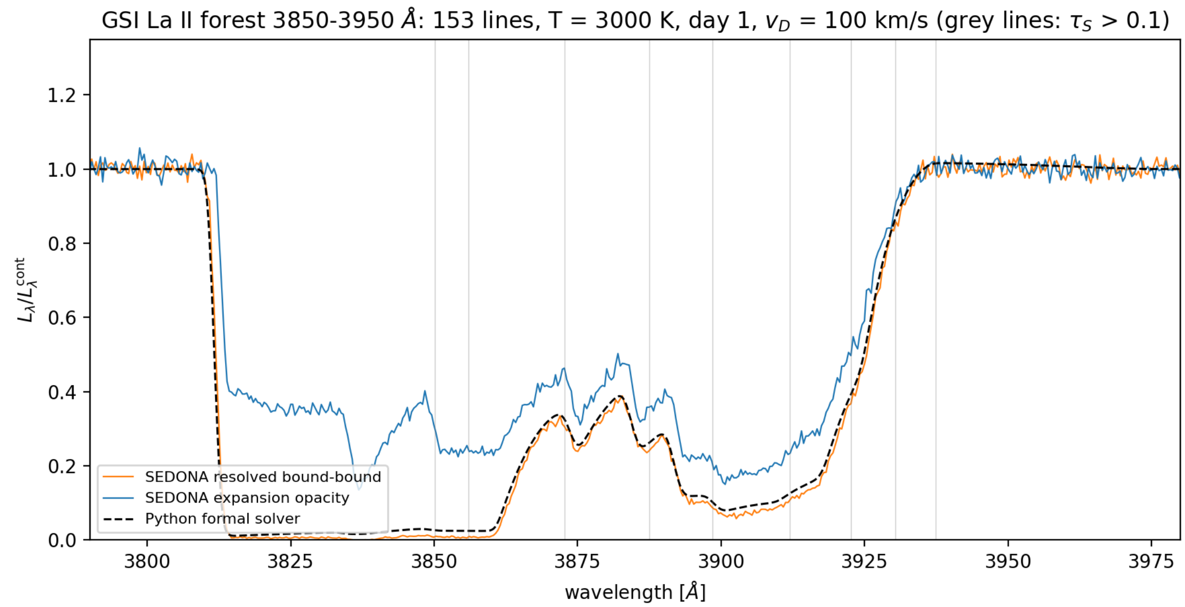

Figure 5: The 3850–3950 Å La II forest (153 calibrated lines, 3 with $\tau_S > 1$, max 5) at $T = 3000$ K, one day after merger. The resolved treatments (orange, black dashed) saturate to black where strong troughs overlap; expansion opacity (blue) never drops below ~ 0.2 because each crossing is capped.

line): real forests are dominated by a few strong lines, precisely where the per-crossing cap is most wrong. That observation prompts the question the next subsection answers—whether the fault lies with the Sobolev approximation or with the way it has been implemented.

5.3 Which approximation is at fault? Separating the two errors

We now put both approximations to the same forest and the same truth. The distinction drawn in §2.5 becomes operational here: the Sobolev approximation *itself* attenuates each line by $e^{-\tau_S}$ with delta-function resonances, while expansion opacity substitutes $1-e^{-\tau_S}$ at every crossing. In this pure-absorption LTE configuration the Sobolev prediction is exact analytics (the p-averaged staircase of Appendix A.5), so it requires no additional transport calculations, and the identical machinery yields the expansion prediction by damping each crossing. A Monte Carlo realization of per-line Sobolev interactions would add only sampling noise to a result available in closed form.

For the La II forest, against SEDONA resolved = 0.3426: Sobolev proper gives 0.3507 (+2.4%) while expansion opacity gives 0.4965 (+44.9%). Across the sweep (Fig. 6):

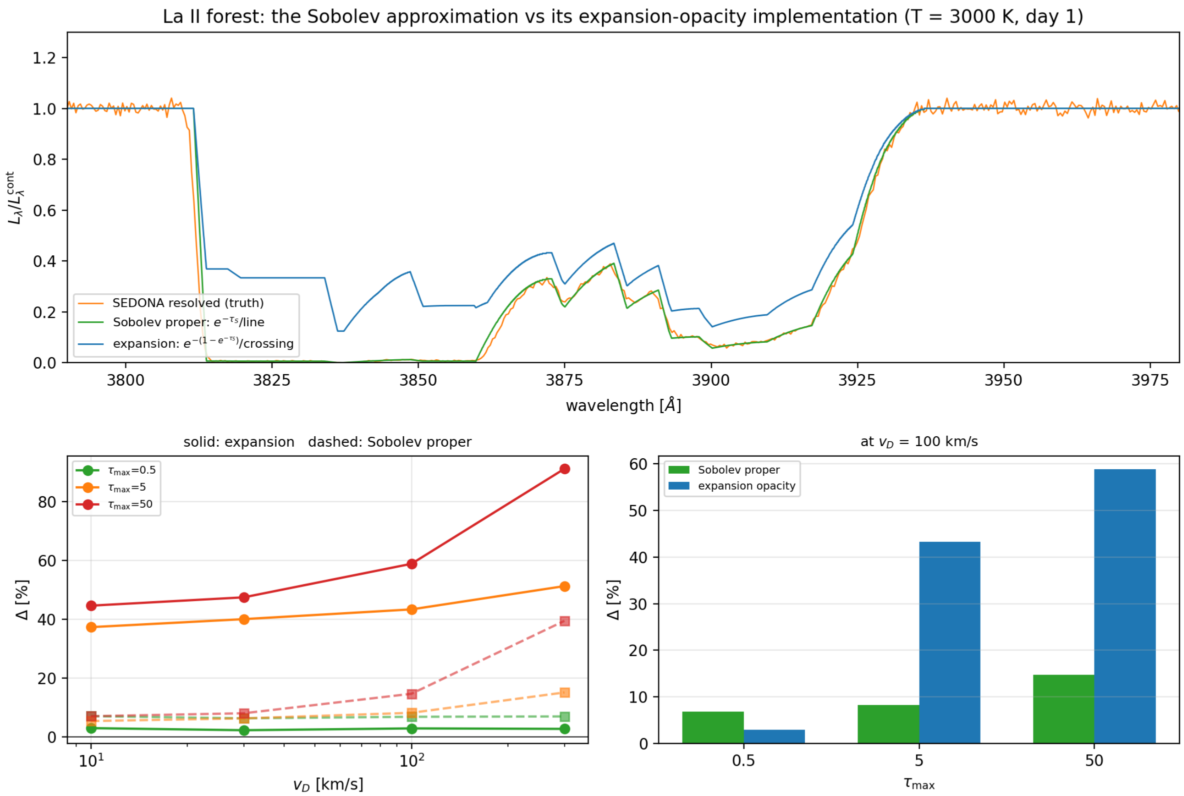

Figure 6: Top: the La II forest under three treatments. The Sobolev approximation proper (green, per-line $e^{-\tau_S}$) tracks the resolved calculation (orange) closely; expansion opacity (blue) sits well above it. Bottom left: both errors versus Doppler width, for three line-strength scalings (solid: expansion; dashed: Sobolev). Bottom right: the two at $v_D = 100\,\mathrm{km\,s^{-1}}$.

$\tau_{\rm max}$	v_D [km/s]	Δ Sobolev	Δ expansion
0.5	10	+7.1%	+3.0%
0.5	300	+7.0%	+2.8%
5	10	+5.4%	+37.3%
5	300	+15.1%	+51.2%
50	10	+7.1%	+44.6%
50	300	+39.4%	+91.1%

Finding F9: the two components of F6 have different owners. *The strength-set floor belongs to expansion opacity alone.* Sobolev proper exhibits no strength floor at all—it is +5–7% at $\tau_{\rm max} = 0.5$, 5 and 50 alike at small v_D—while expansion climbs +3% → +37% → +45%. That floor is the per-crossing cap, not a failure of locality or isolation. *The width-growing term, by contrast, is a genuine Sobolev failure*: a delta-function resonance has no width by construction, so the Sobolev leg is exactly v_D-independent and all of the resolved calculation's width dependence is unmodelled, reaching +39% at $(\tau_{\rm max}, v_D) = (50, 300\,\mathrm{km\,s^{-1}})$.

The headline is therefore sharper than either treatment alone suggests: at kilonova-relevant line widths ($\lesssim 30\,\mathrm{km\,s^{-1}}$) **the Sobolev approximation is accurate to** $\approx$ 5–8%**, while its**

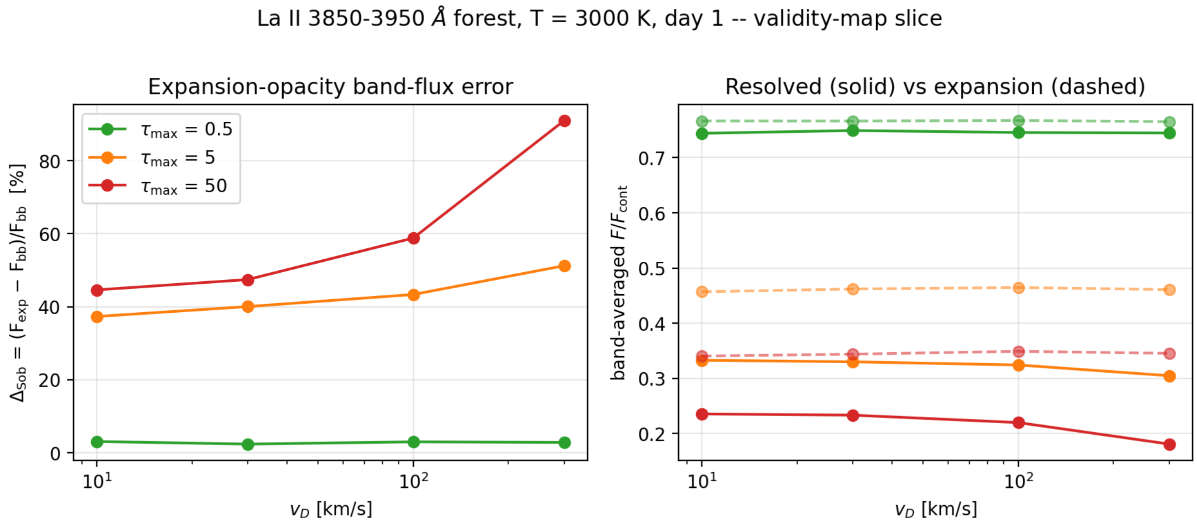

Figure 7: First validity-map slice: $\Delta_{\rm Sob}$ vs Doppler width for three overall line-strength scalings of the La II forest. Right panel: expansion-opacity fluxes (dashed) are blind to v_D, while resolved fluxes (solid) drop as profiles widen.

standard expansion-opacity implementation errs by 40–90%. The literature conflates the two under one name; they are different approximations and they fail for different reasons.

Two caveats. At $\tau_{\rm max} = 0.5$ the SEDONA expansion result (+3.0%) is closer to truth than Sobolev proper (+7.1%), but this is compensating error rather than accuracy: the analytic cap there is +10.1%, and SEDONA's binning smears absorption into inter-line gaps, darkening the spectrum by $\sim 6\%$ and offsetting the cap's over-transmission. And the residual +5–7% Sobolev floor at small v_D is not explained by line strength; the natural candidate is overlap within a Doppler width—the isolation assumption—which the crowding statistics of §5 show is present in these data.

5.4 Where each error lives: strength, width, temperature

The separation just established is a single point in parameter space. We now map where each error lives, by scaling the two quantities the two approximations should respond to differently: the overall line strength, which drives the per-crossing cap, and the Doppler width, which the Sobolev treatment ignores by construction.

Scaling the overall line strength ($\tau_{\rm max} \in \{0.5, 5, 50\}$ by scaling the ion density) and the Doppler width ($v_D \in \{1, 3, 10, 30, 100, 300\}\,{\rm km\,s^{-1}}$) yields the first rows of the validity map (Figs. 7–8; $\Delta_{\rm Sob}$ in percent):

$\tau_{\rm max}\backslash v_D$	1	3	10	30	100	300 km/s
0.5	–	–	+3.0	+2.3	+2.9	+2.8
5	+37.8	+38.1	+37.3	+40.0	+43.3	+51.2
50	–	–	+44.6	+47.4	+58.8	+91.1

Finding F6: the error decomposes into

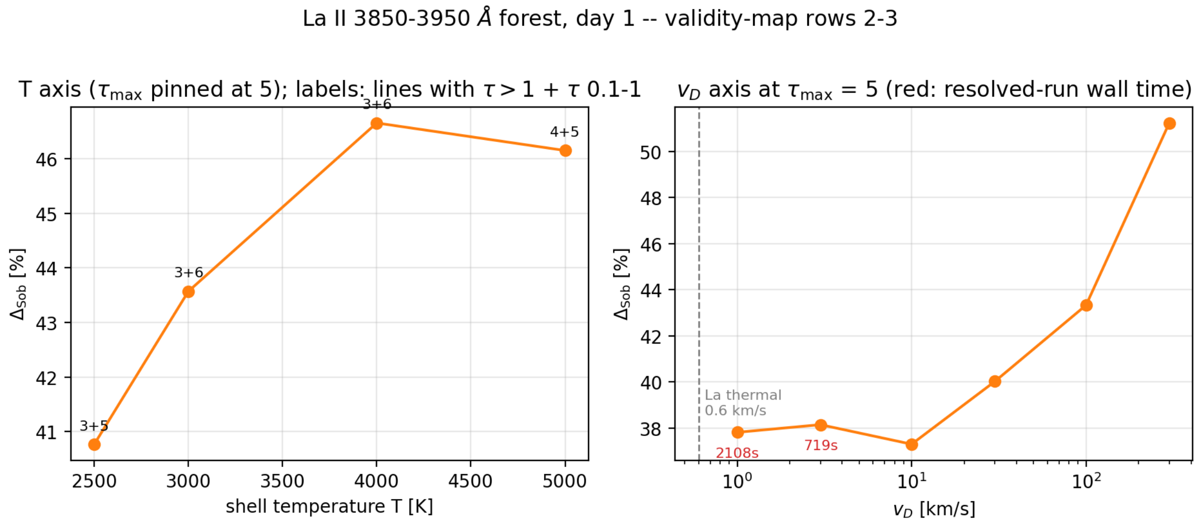

Figure 8: Left: temperature axis at pinned $\tau_{\max} = 5$ (weak dependence; annotations count strong lines). Right: the Doppler-width frontier: Δ_{Sob} is flat at $\sim 38\%$ from $10\,\text{km}\,\text{s}^{-1}$ down to $1\,\text{km}\,\text{s}^{-1}$, essentially the La thermal width; red annotations give the resolved-run wall time.

1. a *strength-set floor*, independent of v_D: the pure per-crossing substitution error. Weak forests are safe at the 3% level; by $\tau_{\max} = 5$ the floor is $\sim 38\%$;

2. a *wing term* growing with v_D: expansion opacity is width-blind (its transmitted flux is flat across a factor 300 in v_D), while genuinely wide lines absorb continuum between resonances.

The frontier measurement at $1\,\text{km}\,\text{s}^{-1}$—within a factor 2 of the actual La thermal width—confirms that the floor survives: *no refinement of line widths rehabilitates expansion opacity in the strong-line regime.*

The temperature axis (Fig. 8, left), with strength pinned, moves Δ_{Sob} only from +41% to +47% across 2500–5000 K: in this window the same few strong lines dominate at all temperatures, so population redistribution is a weak axis (single-ion result; multi-ion mixtures may differ).

5.5 Is the separation universal? A breadth sweep

The measurements above rest on one wavelength window at one epoch. To test their generality we ran a sweep over four windows (4300, 4900, 7000 and 9100 Å, selected automatically by optical-depth richness and deliberately excluding the window used so far), three epochs ($t = 0.5$, 1 and 3 days at fixed ejecta mass, so $\rho \propto t^{-3}$ and $\tau_{\text{S}} \propto t^{-2}$), and three ion mixtures (La II; +Ce II; +Ce III). That is 36 conditions and 72 transport calculations, spanning 71 to 2504 lines per window and realized $\tau_{\max}$ from 0.003 to 34.

	median	max	$\tau_{\max} > 3$	$\tau_{\max} < 0.5$
Δ expansion	+5.1%	**+61.3%**	+20.0%	+0.3%
Δ Sobolev	+3.6%	**+11.3%**	+6.6%	+0.5%

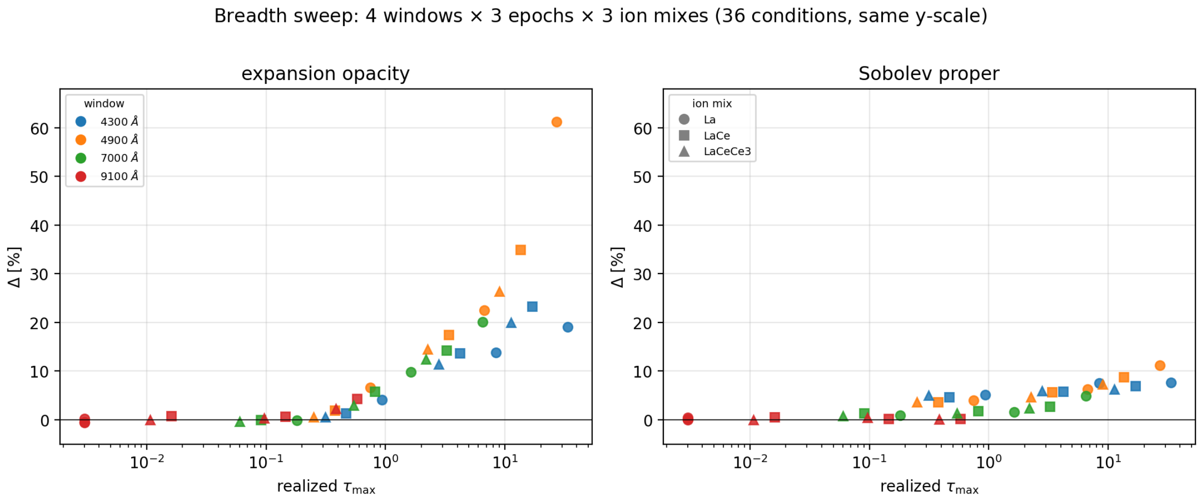

Figure 9: Thirty-six conditions—four wavelength windows (none of them the window of §5.3), three epochs, three ion mixtures—with both panels on the same vertical scale. Every condition collapses onto a single trend in the realized $\tau_{\max}$; expansion opacity reaches +61% while the Sobolev approximation never exceeds +12%.

Finding F10: the separation is universal, and the realized $\tau_{\max}$ is the controlling variable. All 36 conditions fall on one trend (Fig. 9) regardless of window, epoch or composition; those enter only through the optical depths they produce. Both errors vanish together as $\tau_{\max} \to 0$—the weak-line limit both approximations share—and both grow with $\tau_{\max}$, but expansion opacity errs by a median factor 3.5 more than the Sobolev approximation wherever $\tau_{\max} > 1$. This has a practical corollary: the error can be estimated from the strongest line in a band without knowing which ion or epoch produced it.

We note one methodological point that the sweep itself exposed. The 9100 Å window contains no strong La II lines ($\tau_{\max} \simeq 0$) and therefore acts as a free null control: its band flux must be unity. An early pass returned 1.075 instead, tracing to a normalization reference that straddled the final (partial, and unreliable) bin of the transport grid. The bias cancelled exactly in Δ expansion, a same-code differential, but contaminated Δ Sobolev, which compares a Monte Carlo result against an analytic one—the same asymmetry as the emission convention of §3.5. Null controls of this kind cost nothing and are worth including by default.

5.6 Multi-ion blending and the density dependence

One axis remains: what happens when forests become dense enough that resonances stop being resolvable as individual features?

Adding Ce II at equal mass fraction multiplies the line count by 16 and drives strong-line separations down to $8\,\mathrm{km\,s^{-1}}$—genuine sub-Doppler blending. The measured error, however, *falls*, to $\Delta_{\mathrm{Sob}} = +14.6\%$ (Fig. 10).

Finding F7: Δ_{Sob} is non-monotonic in forest density. With enough crossings, even capped per-crossing contributions sum to blackness, so both treatments agree that almost nothing escapes and the relative error is suppressed. The regime that maximizes the error is the *intermediate* one—strong lines that do not yet fully blanket the band—which is exactly where spectral

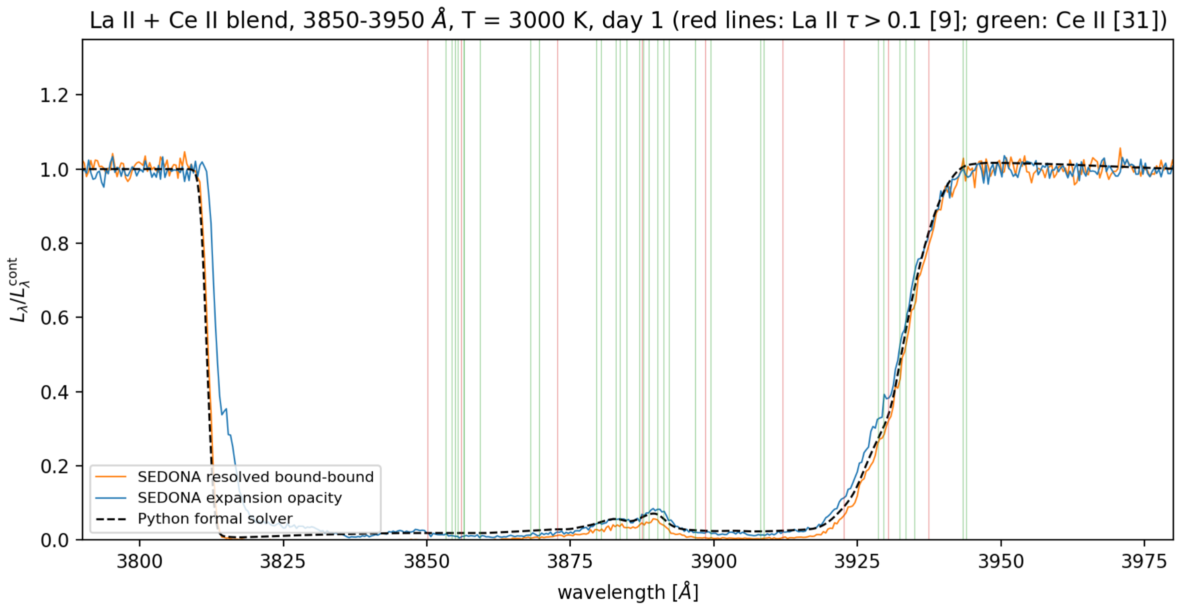

Figure 10: La II + Ce II blend in the same window. Ce floods it with 2376 additional lines (2529 total; strong-line spacings down to $8\,\mathrm{km\,s^{-1}}$), saturating the band to near-black in every treatment.

features form and where composition is read off.

5.7 Computational cost: what the shortcut buys

Why does everyone use the shortcut? Cost. On a standard SEDONA Type Ia example, switching from expansion opacity to the resolved treatment multiplied the run time by 378 (Fig. 11). Our sweep adds a cleaner measurement: in the controlled forest, resolved-run wall time scales as $1/v_D$ (25 s at $100\,\mathrm{km\,s^{-1}}$, 719 s at $3\,\mathrm{km\,s^{-1}}$, 2109 s at $1\,\mathrm{km\,s^{-1}}$), driven entirely by the transport frequency grid—and the expansion run on the same fine grid costs the same (2305 s at $1\,\mathrm{km\,s^{-1}}$). The expense is the grid, not the physics; narrow windows keep resolved calculations affordable, which is the strategy this program uses.

6 Discussion

What the floor means for kilonova modeling. Lanthanide-rich ejecta are the definition of the strong-line regime: Sobolev depths of important lines reach 10^2–10^4 in the line-forming region. Our measurements say that in that regime the expansion-opacity treatment overestimates band-integrated transmitted flux by tens of percent *as a floor*, independent of how finely line profiles are treated. Band fluxes feed directly into inferred ejecta masses and compositions; a systematic of this size and sign (too much blue light escaping) plausibly biases lanthanide-fraction inferences. Quantifying that propagation to observables is future work.

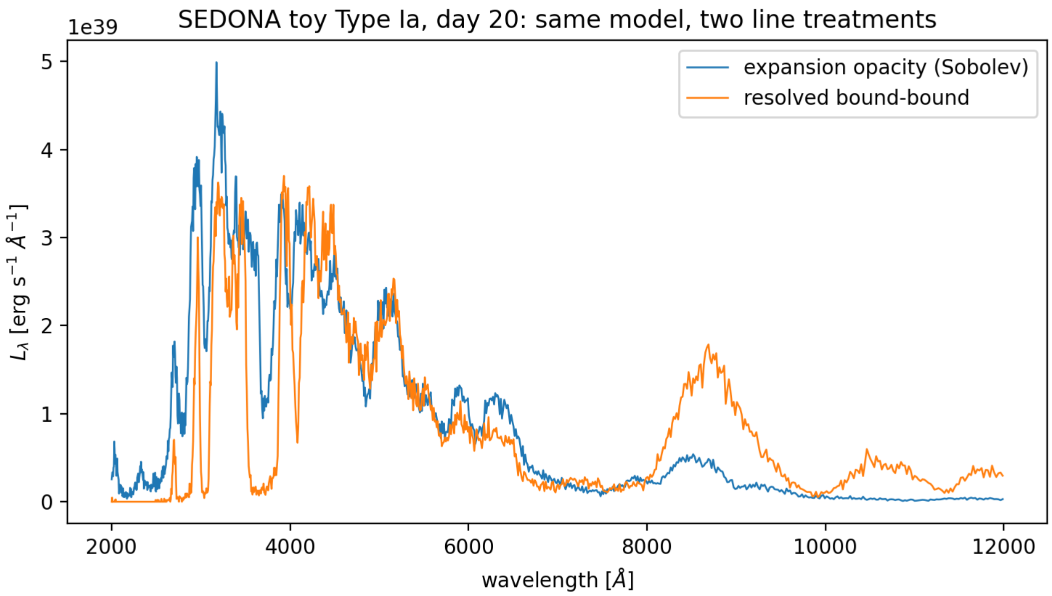

Figure 11: Motivating cost anecdote from a standard SEDONA Type Ia supernova example: the same model runs in 14 s with expansion opacity and 88 min resolved (378×), while redistributing flux from the UV to the near-IR.

What exactly was measured on the Sobolev side. Our Sobolev leg is the approximation in its pure form—$e^{-\tau_S}$ per line, resonances treated as planes—evaluated analytically. That evaluation is *exact* rather than approximate, but only because the configuration is pure absorption without scattering: it is precisely that restriction which makes the p-averaged staircase equal to per-line Sobolev transport. A per-line Sobolev scheme inside a scattering Monte Carlo code—the macroatom treatment of TARDIS-class codes (Kerzendorf and Sim, 2014), say—is a different object, and our numbers should not be read as bounding it; extending the harness there is the natural sequel, and is the regime in which the leg would have to be built rather than computed.

Within that scope the attribution is clean. The per-crossing cap (F4, F5) is absent from the Sobolev leg by construction and present in expansion opacity by construction, and the measured errors follow suit. Of the two Sobolev-intrinsic assumptions, locality is not stressed by our constant-population models—F1 shows it would cancel at leading order even if it were—so the residual few-percent Sobolev error most plausibly belongs to the isolation assumption, i.e. to overlap within a Doppler width. The crowding statistics of §5.1 confirm such overlap exists in these data; quantifying its contribution directly is the first item of future work.

Limitations. Pure absorption with LTE populations (no scattering, fluorescence, or NLTE); fixed temperatures; controlled shell velocities (1000–3000 km s^{-1}) rather than realistic 0.1–0.3c (pending frame-consistent treatment, F3); Doppler widths down to 1 km s^{-1} but not the 0.6 km s^{-1} thermal value; two elements and three ionization stages across four windows and three epochs, but no full lanthanide mixture; Monte Carlo noise $\sim$ 1–2% per band flux. None of these caveats touches the sign or order of magnitude of the central result, which is reproduced independently by analytic theory (Appendix A.4).

Table 1: Findings register.

#	Finding	Where
F1	Symmetric gradients cancel the Sobolev error at leading order; gradient-based diagnostics are conservative	§4
F2	Resolved vs expansion cost: 378× on a standard example	§5
F3	The $\mathcal{O}(v_{\rm bulk}/c)$ offset from comoving Sobolev τ is an artifact of frozen-snapshot integration, not of relativity	§4, App. A.6
F4	Expansion opacity transmits $e^{-(1-e^{-\tau})}$ per crossing, not $e^{-\tau}$	§4
F5	That error is per-resonance: it does not average away with line count	§4
F6	$\Delta_{\rm Sob}$ = strength-set floor + v_D-growing wing term; the floor survives to thermal widths	§5
F7	$\Delta_{\rm Sob}$ is non-monotonic in forest density: maximal for strong sparse forests, suppressed at full line blanketing	§5.6
F8	Solver-vs-MC offsets trace to the thermal-emission convention, not to profiles or resolution; matched, the codes agree to $\sim 1\%$	§3.5
F9	The strength floor is expansion opacity's alone; the width term is a genuine Sobolev failure. Sobolev $\approx$ 5–8% vs 40–90%	§5.3
F10	The separation is universal across windows, epochs and ion mixes; realized $\tau_{\max}$ is the controlling variable	§5.5
F11	Neglect of light-travel-time evolution is a distinct approximation: frozen $\tau/\tau_{\rm S} = (1-\beta)/\gamma$ vs worldline $1/\gamma$	App. A.6

7 Conclusions and next steps

We built and validated a harness that isolates the line-transfer treatment in expanding ejecta—three independent calculations agreeing to $\sim 1\%$ once conventions are matched—and used it to measure, on calibrated lanthanide data, two errors that the literature carries under a single name.

The Sobolev approximation itself is accurate to $\approx$ 5–8% at kilonova-relevant line widths and shows no dependence on line strength; its one real failure is the neglect of finite line width, which matters only for strong lines at large Doppler widths. Its standard expansion-opacity implementation is a different matter: a per-resonance substitution $\tau \to 1 - e^{-\tau}$ gives it a strength-set error floor of 40–90% in band flux that no refinement of line widths removes, because the substitution is not a discretization but part of the formalism. Across 36 conditions spanning four windows, three epochs and three ion mixtures, the separation is universal and controlled by the realized $\tau_{\max}$ alone. The error is largest not in fully blanketed bands, where both treatments agree that little escapes, but in the intermediate regime where strong lines have not yet merged—which is where spectral features form and where composition is read off.

The practical implication for kilonova modelling is that "we use the Sobolev approximation" and "we use expansion opacity" should not be treated as interchangeable statements, and that the second deserves a quantified error budget wherever strong lines are present.

Next: explaining the residual few-percent Sobolev error, whose natural candidate is line overlap within a Doppler width; frame-consistent comparison at realistic ejecta velocities (0.1–0.3c); and extension to scattering and fluorescence, which is where the analytic equivalence exploited here breaks down and a per-line Sobolev *transport* scheme must be built and tested rather than computed in closed form.

Reproducibility

All code, data provenance, executed notebooks, experiment generators, and this manuscript live at https://github.com/zhuygln/revisit_sobolev. Atomic data: GSI Database for Kilonova Radiative Transfer, Zenodo record 19335084 (CC-BY 4.0). Radiative transfer: the public SEDONA release (https://github.com/dnkasen/pubsed). Every figure regenerates from a committed script; the 27-test suite pins the physics of every module.

A Derivations for the non-specialist

A.1 The formal solution and thermal populations

Multiply Eq. (1) by e^{τ} with $\mathrm{d}\tau = \alpha\,\mathrm{d}s$ and integrate: the intensity leaving a slab is

$$I^{\mathrm{out}} = I^{\mathrm{in}} e^{-\tau_{\mathrm{tot}}} + \int_0^{\tau_{\mathrm{tot}}} S(\tau')\, e^{-(\tau_{\mathrm{tot}}-\tau')}\,\mathrm{d}\tau', \qquad S \equiv j/\alpha. \tag{11}$$

The first term is the attenuated input beam; the second adds the gas's own emission, each parcel attenuated by the material still in front of it. The deterministic solver evaluates Eq. (11) numerically along each ray with $S = B_\nu(T)$, accumulating τ by the trapezoidal rule; convergence requires resolving the Doppler width along the path.

In thermal equilibrium the fraction of ions in level i (energy E_i, statistical weight $g_i = 2J_i+1$) is

$$\frac{n_i}{n_{\mathrm{ion}}} = \frac{g_i\, e^{-E_i/k_B T}}{\sum_j g_j\, e^{-E_j/k_B T}}. \tag{12}$$

A subtlety our tests caught: the *fraction per statistical weight* is what decreases monotonically with energy; raw fractions need not (at 3000–5000 K a $g = 7$ level at $1016\,\mathrm{cm}^{-1}$ out-populates the $g = 5$ ground state of La II).

A.2 The Sobolev optical depth

Along a ray toward the observer, the gas at coordinate z moves with line-of-sight velocity z/t (homologous flow), so a photon of observer frequency ν has gas-frame frequency $\nu' = \nu\,(1-z/ct)$ to first order in v/c. Insert the line opacity (Eq. 5) into the optical-depth integral (Eq. 2) and change variables from z to ν' ($\mathrm{d}\nu' = -(\nu/ct)\,\mathrm{d}z \approx -(\nu_0/ct)\,\mathrm{d}z$ near resonance):

$$\tau = \sigma_{\mathrm{cl}} f \int n_l(z)\,\phi(\nu'-\nu_0)\,\mathrm{d}z \;\approx\; \sigma_{\mathrm{cl}} f\, n_l(z_{\mathrm{res}})\,\frac{ct}{\nu_0}\underbrace{\int \phi\,\mathrm{d}\nu'}_{=1} = \sigma_{\mathrm{cl}} f\, n_l\, \lambda_0\, t, \tag{13}$$

which is Eq. (7). The two approximations made are exactly the two Sobolev assumptions: n_l was pulled out of the integral at its resonance-point value (locality), and the profile was integrated as if no other line interfered (isolation).

A.3 Why a symmetric gradient cancels (F1)

Let $n_l(v) = n_0[1 + g(v - v_{\rm res})]$ with g odd about the resonance (e.g. tanh centered there). The resolved integral weights n_l by the even profile ϕ; the odd term integrates to zero, so the first-order error vanishes. The next term is set by the curvature $n_l''(v_{\rm res})$—which for a centered tanh *also* vanishes (inflection point). Only with the resonance off-center does the curvature term survive, giving $E_{\rm Sob} \simeq \frac{1}{2}|n_l''/n_l|\,\sigma_v^2 \propto \epsilon^2$, the law measured in Fig. 1.

A.4 The expansion-opacity cap (F4)

Integrate the binned opacity of Eq. (8) along the stretch of path over which the moving gas keeps one line inside one frequency bin. The Doppler sweep rate ties path length to bin width, $\Delta z = ct\,\Delta\nu_{\rm bin}/\nu$, so the effective optical depth contributed by line k is

$$\tau_k^{\rm eff} = \alpha^{\rm exp}\,\Delta z = \frac{1}{ct}\,\frac{\nu_k}{\Delta\nu_{\rm bin}}\left(1 - e^{-\tau_{{\rm S},k}}\right) \times \frac{ct\,\Delta\nu_{\rm bin}}{\nu_k} = 1 - e^{-\tau_{{\rm S},k}}. \tag{14}$$

The bin width cancels: independent of resolution, each resonance crossing contributes at most $\tau^{\rm eff} = 1$, and the transmitted fraction is $e^{-(1-e^{-\tau_{\rm S}})}$ per line—Eq. (9), the measured 0.421 at $\tau_{\rm S} = 2$. Note this is *not* a discretization error: it is built into the formalism and no bin refinement removes it. It disappears only when $\tau_{\rm S} \ll 1$.

A.5 Averaging over the source disk (F5's technical companion)

A resonance plane at height $z_{\rm res}$ in front of a core of radius $r_{\rm core}$ is crossed by the core ray at impact parameter p only if $\sqrt{r_{\rm core}^2 - p^2} < z_{\rm res} < \sqrt{r_{\rm out}^2 - p^2}$. The observed attenuation is therefore

$$\frac{F}{F_{\rm cont}}(\nu) = \frac{2}{r_{\rm core}^2}\int_0^{r_{\rm core}} p\,\exp\Big[-\sum_{k\,{\rm crossed}(p,\nu)} \tau_k\Big]\,{\rm d}p, \tag{15}$$

not $e^{-\sum\tau}$ with a global crossing criterion: planes at $z_{\rm res} < r_{\rm core}$ shadow only part of the disk. Ignoring this (∼plane counting) predicts measurably too-shallow troughs in the ladder experiment.

A.6 Frozen snapshot versus photon worldline: what F3 really was

This appendix was wrong in the previous draft, in an instructive way. We keep the faulty step visible because it is easy to make and hard to see.

Appendix A.2 derived $\tau_{\rm S}$ to first order in v/c. Kilonova ejecta move at 0.1–0.3c, and we measured (Finding F3) that a resonance at $z_{\rm res}/ct = 7.7\%$ gave a trough matching $\exp[-\tau_{\rm S}(1 - z_{\rm res}/ct)]$ to four digits rather than $\exp(-\tau_{\rm S})$. The natural conclusion—which we drew, and published—was that this is the leading relativistic correction and that it is $\mathcal{O}(v/c)$. It is not.

The faulty step. That derivation differentiated the Doppler factor $D = \gamma(1-b)$ *at fixed* t, i.e. it pushed a photon through a frozen snapshot of the ejecta. But a photon takes time to cross a resonance, and in homologous flow $\boldsymbol{\beta} = \mathbf{r}/(ct)$ depends on the age as well as the position. Along the photon worldline $\mathbf{r}(t) = \mathbf{r}_0 + c\hat{\mathbf{n}}(t - t_0)$,

$$\frac{{\rm d}\boldsymbol{\beta}}{{\rm d}t} = \frac{\hat{\mathbf{n}} - \boldsymbol{\beta}}{t} \qquad \Longrightarrow \qquad \frac{{\rm d}b}{{\rm d}t} = \frac{1-b}{t}, \tag{16}$$

whereas the frozen calculation uses $\mathrm{d}b/\mathrm{d}z = 1/(ct)$. The two differ by a factor $(1-b)$—first order in β, the same order as the effect being computed. The time term is not negligible even though the crossing takes a negligible fraction of t, because what matters is the *gradient*, not the elapsed time.

The correct law. With $D = \gamma(1-b)$ one finds $\mathrm{d}D/\mathrm{d}t = -D\gamma^2(1-b)/t$. Using the opacity transformation $\chi_{\mathrm{lab}} = D\chi'$ (from the invariance of $\nu\chi_\nu$) and the resonance condition $D\nu = \nu_0$, the profile integral collapses to $\tau = D\chi' c/|\mathrm{d}\nu'/\mathrm{d}t|$, and the factors of $\gamma(1-b)$ cancel down to a single Lorentz factor:

$$\boxed{\frac{\tau_{\mathrm{rel}}}{\tau_{\mathrm{S}}} = \frac{1}{\gamma} = \sqrt{1-\beta^2}} \tag{17}$$

with τ_{S} evaluated using the density and ejecta age local to the resonance, which makes the statement independent of how the calculation is anchored in time. **There is no first-order term**: $1/\gamma = 1 - \frac{1}{2}\beta^2 + \mathcal{O}(\beta^4)$. The frozen calculation instead gives, for radial rays,

$$\frac{\tau_{\mathrm{frozen}}}{\tau_{\mathrm{S}}} = \frac{1-\beta}{\gamma}. \tag{18}$$

Numerical confirmation. Integrating the resolved line opacity directly along the path—no Sobolev assumption anywhere—reproduces both expressions to five decimal places, with the ejecta age held fixed or allowed to advance:

β	frozen τ/τ_{S}	worldline τ/τ_{S}	$1/\gamma$
0.05	0.94881	0.99875	0.99875
0.10	0.89549	0.99499	0.99499
0.20	0.78384	0.97980	0.97980
0.30	0.66776	0.95394	0.95394

So the relativistic correction at $\beta = 0.1$, 0.2, 0.3 is 0.5%, 2.0%, 4.6%—not the 10%, 22%, 33% that the frozen law implies and that the previous draft of this appendix claimed.

Which problem does the Monte Carlo code solve? It is natural to assume a Monte Carlo transport code supplies the time-dependent answer, and we briefly did. It does not, in the configuration used here. Sweeping a single line's resonance velocity from $\beta = 0.001$ to 0.34 and comparing the measured $\tau_{\mathrm{eff}} = -\ln(F/F_{\mathrm{cont}})$ against both laws gives an RMS deviation over $\beta_{\mathrm{res}} \gtrsim 0.1$ of 0.024 from the frozen law and 0.55 from the worldline law—a factor 23, and at $\beta_{\mathrm{res}} = 0.34$ the measurement is 0.613 against a frozen prediction of 0.618 and a worldline prediction of 1.43. SEDONA run with `transport_steady_iterate` iterates the radiation field on a fixed grid at fixed epoch; its source disables hydrodynamics in that mode and reads no time-stepping parameters. It is a frozen-snapshot calculation.

This matters for the harness. Cross-code comparison requires matching the *transport treatment* as well as the thermal-emission convention (F8) and the spectral normalization (§5.5): a third convention, and the one with the largest lever arm at high velocity. Like-for-like against a frozen SEDONA means running the solver in its frozen mode. At the shell velocities used for the measurements in this paper ($\beta \leq 0.01$) the two laws differ by $\lesssim 2\%$ in τ, so this choice does not affect any result reported here; at kilonova velocities it would dominate.

What F3 is, finally. Not a frame ambiguity (our first reading), not the leading relativistic correction (our second, published, reading), but an artifact of integrating a frozen snapshot—a problem the photon does not solve. The distinction earns its own entry in Table 1 as F11, because "neglect of light-travel-time evolution" is a separate approximation from the Sobolev localization, from the independent-line assumption, and from expansion-opacity binning. It is a fourth member of the family this paper is about. Its practical consequence for the measurements reported here is that at our shell velocities ($\beta \leq 0.01$) the correct correction is 5×10^{-5}: negligible, and therefore *not* an explanation for the few-percent Sobolev residual of §5.3.

For the general case the resonance locus is not a plane—γ depends on r, not z—and one needs the common-direction and common-point frequency surfaces of Jeffery (1993). Our controlled experiments keep the emitting core small compared with the resonance radii, so all core rays are nearly radial and the plane picture holds.

Appendix A.2 derived τ_S to first order in v/c. Kilonova ejecta move at 0.1–0.3c, so the neglected terms are not small, and we found empirically (Finding F3) that a resonance at $z_{\rm res}/ct = 7.7\%$ produced a trough matching $\exp[-\tau_S(1 - z_{\rm res}/ct)]$ to four digits rather than $\exp(-\tau_S)$. We originally recorded that as a frame *ambiguity* to be avoided by keeping resonances slow. The derivation below shows it is not an ambiguity: it is the leading relativistic correction, and it is the nonrelativistic τ_S that is wrong.

References

B. P. Abbott et al. Multi-messenger observations of a binary neutron star merger. *The Astrophysical Journal Letters*, 848(2):L12, 2017. doi: 10.3847/2041-8213/aa91c9.

J. Barnes and D. Kasen. Effect of a high opacity on the light curves of radioactively powered transients from compact object mergers. *The Astrophysical Journal*, 775(1):18, 2013. doi: 10.1088/0004-637X/775/1/18.

J. I. Castor. Spectral line formation in Wolf–Rayet envelopes. *Monthly Notices of the Royal Astronomical Society*, 149(2):111–127, 1970. doi: 10.1093/mnras/149.2.111.

R. G. Eastman and P. A. Pinto. Spectrum formation in supernovae: numerical techniques. *The Astrophysical Journal*, 412:731–751, 1993. doi: 10.1086/172957.

A. Flörs et al. GSI database for kilonova radiative transfer. Zenodo, 2026a. Record 19335084; concept DOI 10.5281/zenodo.15835360.

Andreas Flörs, Ricardo Ferreira da Silva, José Pires Marques, Jorge Miguel Sampaio, and Gabriel Martínez-Pinedo. Calibrated lanthanide atomic data for kilonova radiative transfer: Atomic structure and opacities. *Physical Review D*, 113(6):063041, 2026b. doi: 10.1103/jxqw-7ynk.

D. J. Jeffery. The relativistic Sobolev method applied to homologously expanding atmospheres. *The Astrophysical Journal*, 415:734, 1993. doi: 10.1086/173196.

A. H. Karp, G. Lasher, K. L. Chan, and E. E. Salpeter. The opacity of expanding media: the effect of spectral lines. *The Astrophysical Journal*, 214:161–178, 1977. doi: 10.1086/155241.

D. Kasen, R. C. Thomas, and P. Nugent. Time-dependent Monte Carlo radiative transfer calculations for three-dimensional supernova spectra, light curves, and polarization. *The Astrophysical Journal*, 651(1):366–380, 2006. doi: 10.1086/506190.

D. Kasen, B. Metzger, J. Barnes, E. Quataert, and E. Ramirez-Ruiz. Origin of the heavy elements in binary neutron-star mergers from a gravitational-wave event. *Nature*, 551(7678): 80–84, 2017. doi: 10.1038/nature24453.

W. E. Kerzendorf and S. A. Sim. A spectral synthesis code for rapid modelling of supernovae. *Monthly Notices of the Royal Astronomical Society*, 440(1):387–404, 2014. doi: 10.1093/mnras/stu055.

B. D. Metzger. Kilonovae. *Living Reviews in Relativity*, 23:1, 2020. doi: 10.1007/s41114-019-0024-0.

N. Roth and D. Kasen. Monte Carlo radiation-hydrodynamics with implicit methods. *The Astrophysical Journal Supplement Series*, 217(1):9, 2015. doi: 10.1088/0067-0049/217/1/9.

V. V. Sobolev. *Moving Envelopes of Stars*, volume 10 of *Harvard Books on Astronomy*. Harvard University Press, Cambridge, MA, 1960. doi: 10.4159/harvard.9780674864658. Translated by S. Gaposchkin from the 1947 Russian original.

M. Tanaka, D. Kato, G. Gaigalas, and K. Kawaguchi. Systematic opacity calculations for kilonovae. *Monthly Notices of the Royal Astronomical Society*, 496(2):1369–1392, 2020. doi: 10.1093/mnras/staa1576.

M. Tanaka et al. Systematic opacity calculations for kilonovae – II. improved atomic data for singly ionized lanthanides. *Monthly Notices of the Royal Astronomical Society*, 2025. doi: 10.1093/mnras/stae2504.